FYS5419/9419, January 29, 2025

FYS5419/9419 January 29, 2025

$$\sigma_z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

$$\sigma_z \begin{bmatrix} 1 \\ 0 \end{bmatrix} = +1 \underbrace{\begin{bmatrix} 1 \\ 0 \end{bmatrix}}$$

eigenbasis for $\sigma_z = Z$

$\begin{bmatrix} 1 \\ 0 \end{bmatrix} = |0\rangle$ = north pole on Block sphere

$$\sigma_z \begin{bmatrix} 0 \\ 1 \end{bmatrix} = -1 \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$, south pole

$$\hat{\sigma}_x = \hat{X} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$$\hat{X}\begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}\begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$|0\rangle \to |1\rangle$$

$|0\rangle$ and $|1\rangle$ are not eigen states of X (Y)

Measurements

$$\hat{A}\,|\psi_a\rangle = \lambda_a\,|\psi_a\rangle$$

projection operators

$$\{ \hat{P}_0, \hat{P}_1, \dots \hat{P}_{m-1} \}$$

$$m \leq n$$

$$\mathbb{1} = \sum_{i=0}^{n-1} \hat{P}_i$$

Example : $|0\rangle$ and $|1\rangle$

$$\mathbb{1} = \sum_{i=0}^{1} |i\rangle\langle i| = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\mathrm{Prob}(\lambda_i) = |P_i|\psi_a\rangle|^2$$

$$= \langle \psi_a | P_i^{+} P_i | \psi_a \rangle$$

$$(P_i^{+} = P_i \wedge P_i^2 = P_i)$$

$$= \langle \psi_a | P_i | \psi_a \rangle$$

Total probability for all measurements

$$\sum_{i=0}^{n-1} \mathrm{Prob}(\lambda_i) = 1$$

After measurement, the post measurement, normalized state after the outcome i:

$$\frac{\hat{P}_i |\psi_a\rangle}{\sqrt{\langle\psi_a|\hat{P}_i|\psi_a\rangle}}$$

Example 1

$$\hat{P}_0 = |0\rangle\langle 0| \quad \hat{P}_1 = |1\rangle\langle 1|$$

$$\sum \hat{P}_i^{\dagger} \hat{P}_i = \mathbb{1}$$

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

$$\mathrm{Prob}(\psi_0) = \langle\psi|\underline{\hat{P}_0^{\dagger}\hat{P}_0}|\psi\rangle$$

$$= \left(\alpha^*\langle 0| + \beta^*\langle 1|\right)|0\rangle\langle 0|$$

$$\times\left(\alpha|0\rangle + \beta|1\rangle\right)$$

$$= |\alpha|^2$$

$$|\psi_0'\rangle = \frac{\hat{P}_0|\psi\rangle}{\sqrt{\langle\psi|P_0|\psi\rangle}} = \frac{\alpha}{|\alpha|}|0\rangle$$

$$|\psi_1'\rangle = \frac{\beta}{|\beta|}|1\rangle$$

Example 2

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

$$\text{Prob}(\psi_0) = \langle\psi|\hat{P}_0^{\dagger}P_0|\psi\rangle = |\alpha|^2$$

Density matrix/operator

$$\hat{\rho}_\psi = |\psi\rangle\langle\psi|$$

$$= \begin{bmatrix} |\alpha|^2 & \alpha\beta^* \\ \alpha^*\beta & |\beta|^2 \end{bmatrix}$$

$$\hat{P}_0^\dagger \hat{P}_0 = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} = \hat{P}_0$$

$$\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} |\alpha|^2 & \alpha\beta^* \\ \alpha^*\beta & |\beta|^2 \end{bmatrix} = \hat{P}_0 \hat{\rho}_\psi$$

$$=$$

$$= \begin{bmatrix} |\alpha|^2 & \alpha\beta^* \\ 0 & 0 \end{bmatrix}$$

$$\Longrightarrow$$

$$\text{prob}(\psi(0)) = \text{Tr}\left[\hat{P}_0^\dagger \hat{P}_0 \hat{\rho}_\psi\right]$$

$$= |\alpha|^2$$

in general

$$\text{prob}(x) = \text{Tr}\left[\overbrace{\hat{P}_x^\dagger \underbrace{\hat{P}_x}_{|x\rangle\langle x|}}^{\hat{M}_x} \hat{\rho}\right]$$

$$|\psi\rangle = \sum_i c_i |i\rangle$$

$$\hat{\rho}_\psi = \sum_i |c_i|^2 |i\rangle\langle i|$$

$|c_i|^2 = p_i$ = probability of being in state $|i\rangle$

$$\boxed{\hat{\rho}_\psi = \sum_i p_i |i\rangle\langle i|}$$

spectral decomposition

Example: Bell states

$$|\phi^+\rangle = \frac{1}{\sqrt{2}}\left[|00\rangle + |11\rangle\right]$$

$$|00\rangle = |0\rangle_A \otimes |0\rangle_B = \begin{bmatrix}1\\0\end{bmatrix} \otimes \begin{bmatrix}1\\0\end{bmatrix}$$

$$= \begin{bmatrix}1\\0\\0\\0\end{bmatrix}$$

$$|11\rangle = |1\rangle_A \otimes |1\rangle_B = \begin{bmatrix}0\\0\\0\\1\end{bmatrix}$$

$$|\phi^-\rangle = \frac{1}{\sqrt{2}}\left[|00\rangle - |11\rangle\right]$$

$$|\psi^+\rangle = \frac{1}{\sqrt{2}}\left[|10\rangle + |01\rangle\right]$$

$$|\psi^-\rangle = \frac{1}{\sqrt{2}}\left[|10\rangle - |01\rangle\right]$$

$$\rho_{\phi^+} = \frac{1}{2}\left[(|00\rangle + |11\rangle)(\langle 00| + \langle 11|)\right]$$

pure state

$$\left(\begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \end{bmatrix} \right) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\hat{\rho}_{\phi^+} = |\phi^+\rangle\langle\phi^+| = \frac{1}{2}\begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 \end{bmatrix} = \hat{\rho}$$

$$\rho^2 = \rho \Rightarrow \mathrm{Tr}[\rho^2] = \mathrm{Tr}[\rho] = \underline{1}$$

Measurements :

$$|00\rangle = |0\rangle_A \otimes |0\rangle_B$$

$$\hat{P}_0 = \underbrace{|0\rangle_A \langle 0|_A}_{2\times 2} \otimes \underbrace{\mathbb{I}_B}_{2\times 2}$$

$$\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \otimes \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\mathrm{Prob}(\phi^+(0)) = {}_{AB}\langle \phi^+ | \hat{P}_0 | \phi^+ \rangle$$

$$= \frac{1}{2} \begin{bmatrix} 1 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$= 1/2$$

$$\mathrm{Prob}(\phi^+(1)) = \frac{1}{2}$$

$\hat{\rho}_{AB}$ what is the density matrix for $\hat{\rho}_A$ or $\hat{\rho}_B$?

$$\hat{\rho}_A = \mathrm{Tr}_B \left(\hat{\rho}_{AB} \right)$$

$$\hat{\rho}_B = \mathrm{Tr}_A \left(\hat{\rho}_{AB} \right)$$

($p(x,y) \rightarrow$ marginal prob

$$p(x) = \int_{y \in D} dy \, p(x,y)$$

Bell state ϕ^+

$$\hat{\rho}_{AB} = \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 \end{bmatrix} \quad \mathbb{R}^{4\times 4}$$

$$\hat{\rho}_A = \mathrm{Tr}_B(\hat{\rho}_{AB}) = \langle 0_B | \rho_{AB} | 0_B \rangle$$

$$+ \langle 1_B | \rho_{AB} | 1_B \rangle$$

$$|0_A\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \otimes \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}$$

4×2

$$\begin{array}{ccc} 2\times 4 & 4\times 4 & 4\times 2 \\ \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix} \times & \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 \end{bmatrix} & \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \\ 0 & 0 \end{bmatrix} \end{array}$$

$$\rho_{AB} \implies$$

$$\hat{\rho}_A = \frac{1}{2} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \frac{1}{2} \mathbb{1} = \hat{\rho}_B$$

$$\mathrm{Tr}\left[\hat{\rho}_A^2\right] = \frac{1}{2} < 1$$

Mixed state

$$\hat{\rho}_A = \hat{\rho}_B = \frac{|0\rangle\langle 0| + |1\rangle\langle 1|}{2}$$

mixed state

$\mathrm{Tr}\,[\rho^2] = 1$ pure state

$\mathrm{Tr}\,[\rho^2] < 1$ mixed state